

Q 1

ACE

Molly owns three brown caps and four red caps. She selects one of the caps for herself at random and another cap at random for two friends. What is the probability of the three caps being red?

- (A) $\frac{4}{35}$
(B) $\frac{64}{343}$

- (C) $\frac{4}{7}$
(D) $\frac{31}{35}$

$$P(RRR) = \frac{4}{7} \times \frac{3}{6} \times \frac{2}{5} = \frac{4}{35} \quad A$$

Q 2

ACE

In a business the employees are 55% male and 45% female. Two employees are selected at random. What is the probability that both are male?

- (A) 0.2025
(B) 0.2475

- (C) 0.3025
(D) 0.5555

$$P(MM) = 0.55 \times 0.55 = 0.3025 \quad C$$

Q 3

ACE

A bag contains two blue balls, one red ball, and one white ball. Sandra selects one ball from the bag and keeps it hidden. She then selects a second ball, also keeping it hidden.

- (i) Find the probability that both the selected balls are blue.
(ii) Find the probability that at least one of the selected balls is blue.
(iii) Sandra drops one of the selected balls and she sees that it is blue. What is the probability that the ball that is hidden is also blue?

15(b) (i) $P(BB) = \frac{2}{4} \times \frac{1}{3} = \frac{1}{6}$

15(b) (ii) Probability of at least one blue ball is every outcome except selecting red and white balls.
 $P(\text{at least 1 blue}) = 1 - P(RW + WR) = 1 - \left(\frac{1}{4} \times \frac{1}{3} + \frac{1}{4} \times \frac{1}{3}\right) = \frac{5}{6}$

15(b) (iii) If one of the selected balls is blue, there are 5 possible outcomes only one of which will yield another blue.

Sample space = {BB, BR, BW, RB, WB}
 $P(E) = \frac{1}{5}$

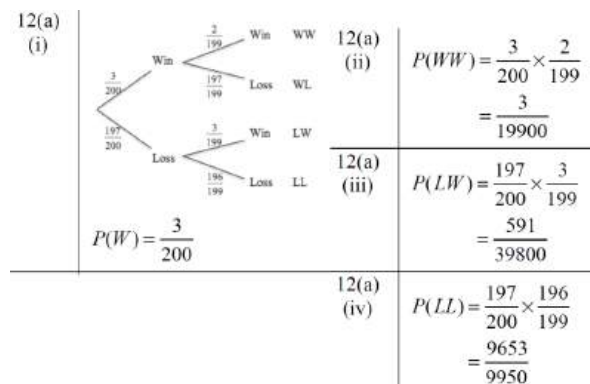
Q 4

ACE

There are 200 tickets sold in a raffle with only two prizes. These tickets are placed in a bag and two are drawn, one at a time. Once a ticket is drawn it is not placed back in the bag. One boy bought 3 tickets.

- (i) What is the probability he wins first prize?
(ii) What is the probability he wins both prizes?
(iii) What is the probability he wins second prize but not first prize?
(iv) What is the probability he does not win a prize?

1
1
1
1



Q 5

ACE

In a school the student population is 45% male and 55% female. Two students are selected at random to represent their school.

- (i) What is the probability that both are female? Answer correct to 2 decimal places.
(ii) What is the probability that one is female and the other is male? Answer correct to 2 decimal places.
(iii) What is the probability that neither student is female? Answer correct to 2 decimal places.

1
2
1

5(b) (i) $P(FF) = 0.55 \times 0.55 = 0.3025$

5(b) (ii) $P(MF \text{ or } FM) = 0.45 \times 0.55 + 0.55 \times 0.45 = 0.495$

5(b) (iii) $P(MM) = 0.45 \times 0.45 = 0.2025$





Q 6

The chance of a fisherperson catching a legal length fish is 4 in 5. If three fish are caught at random, what is the probability that:

ARC

- | | | |
|-------|----------------------------------|---|
| (i) | none are of legal length. | 1 |
| (ii) | all are of legal length. | 1 |
| (iii) | exactly one is of legal length. | 1 |
| (iv) | exactly two are of legal length. | 1 |

a) (i) $\frac{1}{5} \times \frac{1}{5} \times \frac{1}{5} = \frac{1}{125}$

(ii) $\frac{4}{5} \times \frac{4}{5} \times \frac{4}{5} = \frac{64}{125}$

(iii) $\frac{4}{5} \times \frac{1}{5} \times \frac{1}{5} \times 3 = \frac{12}{125}$

(iv) $\frac{4}{5} \times \frac{4}{5} \times \frac{1}{5} \times 3 = \frac{48}{125}$

Q 7

A bag contains 6 white socks and 4 black socks. When you select a sock, you cannot tell what colour it is until it is removed from the bag. Two socks are selected from the bag.

ARC

- | | | |
|-------|--|---|
| (i) | What is the probability that the first sock is white? | 1 |
| (ii) | What is the probability that a pair is selected? | 2 |
| (iii) | If four socks are selected, what is the probability that they are all black? | 1 |

(b)(i) $P(\text{white}) = \frac{6}{10} = \frac{3}{5}$

(ii) $P(\text{pair}) = p(\text{ww}) + p(\text{bb})$
 $= \frac{3}{5} \times \frac{5}{9} + \frac{2}{5} \times \frac{3}{9} = \frac{15}{45} + \frac{6}{45}$
 $= \frac{21}{45} = \frac{7}{15}$

(iii) $P(\text{bbbb}) = \frac{4}{10} \times \frac{3}{9} \times \frac{2}{8} \times \frac{1}{7} = \frac{1}{210}$

Q 8

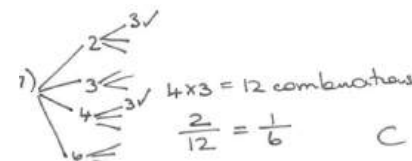
Two-digit numbers are formed from the digits 2, 3, 4, 6 with no repetition of digits allowed.

A two-digit number is then selected at random.

BARKER

What is the probability that the number is prime?

- (A) $\frac{1}{12}$ (B) $\frac{1}{8}$ (C) $\frac{1}{6}$ (D) $\frac{5}{12}$



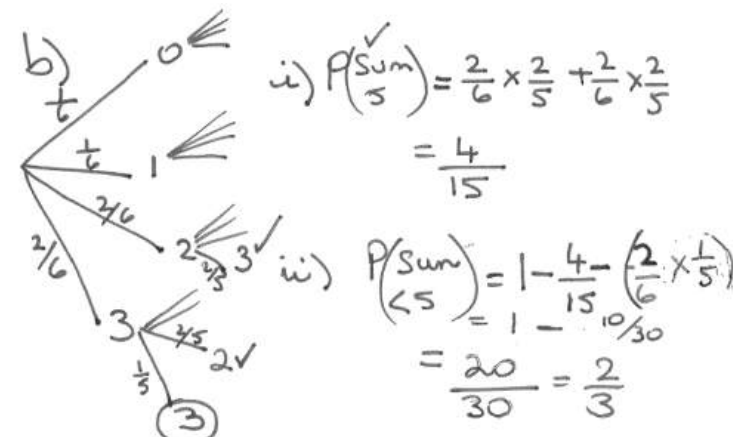
Q 9

A deck contains six cards labelled the numbers 0, 1, 2, 2, 3 and 3 respectively.

Ann draws two cards at random without replacement.

BARKER

- | | | |
|------|--|---|
| (i) | Find the probability that the sum of the two cards equals 5 | 1 |
| (ii) | Find the probability that the sum of the two cards is less than five | 2 |





Q10

BARKER

The probability of gaining relief from a particular illness with antibiotic A is $\frac{7}{10}$,
whilst with antibiotic B, the probability of gaining relief is $\frac{3}{4}$.

One cold sufferer takes A, whilst another takes B.

What is the probability that only one cold sufferer gains relief?

$$= P(R_A N_B) + P(N_A R_B)$$

$$= \frac{7}{10} \times \frac{1}{4} + \frac{3}{10} \times \frac{3}{4}$$

$$= \frac{7}{40} + \frac{9}{40} = \frac{16}{40} = \frac{2}{5}$$

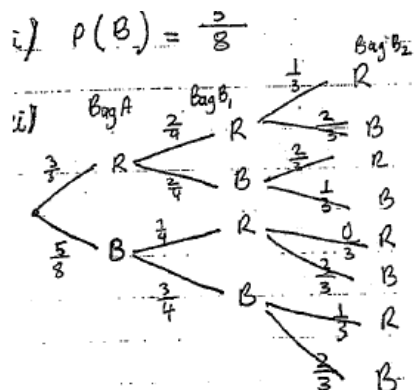
Q11

BARKER

On a table there are two bags A and B. Bag A contains three red and five blue marbles,
whilst bag B contains one red and two blue marbles.

First Ann draws one marble from bag A at random and puts it in bag B.
Then Ann draws two marbles from bag B at random without replacement.

- What is the probability that a marble drawn from bag A is blue? 1
- Using a tree diagram, or otherwise, find the probability that both marbles drawn from bag B are blue? 3



(both Blue from Bag B.)

$$P(RBB) + P(BBB)$$

$$\frac{3}{8} \times \frac{2}{4} \times \frac{1}{3} + \frac{5}{8} \times \frac{3}{4} \times \frac{2}{3}$$

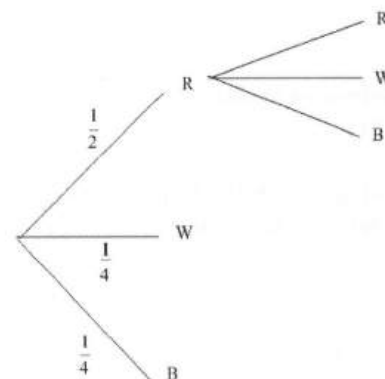
$$= \frac{3}{8}$$

Q12

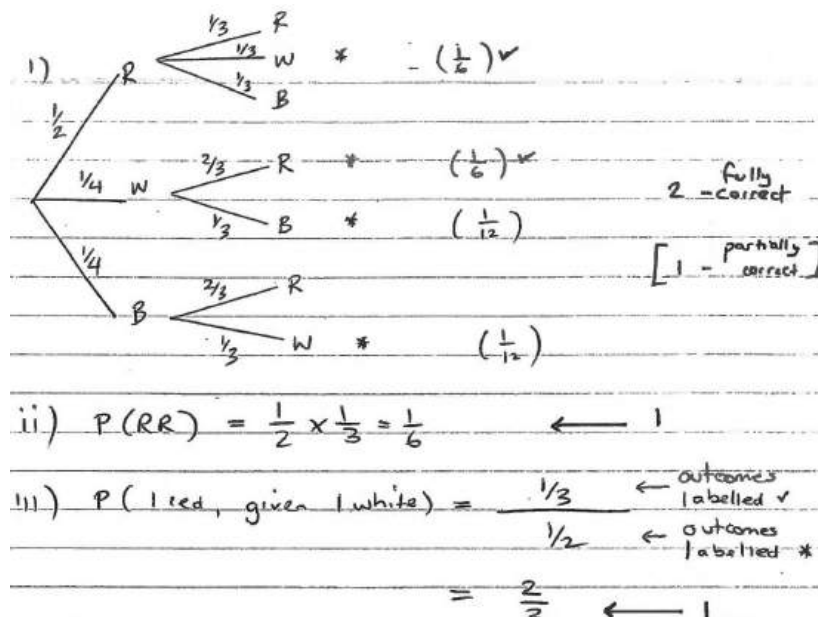
BAULKHAM

A jar contains 2 red, 1 black and 1 white marble only. Ken randomly selects two marbles in succession (without replacement), and places them in a bag.

- Copy and complete the probability tree to show all possible outcomes. 2



- Find the probability that both of the selected marbles are red. 1
- If one of the marbles falls out of the bag and we see that it is white, what is the probability that the marble still in the bag is red? 1





Q13

BAULKHAM

Rita has just been shopping and purchased 3 cans of baked beans and 2 cans of spaghetti. While she is on the phone her little brother removes all the labels from all the cans so that they all look alike now. Her brother wants baked beans for lunch. Rita decides to open only two cans. She selects the two cans at random.

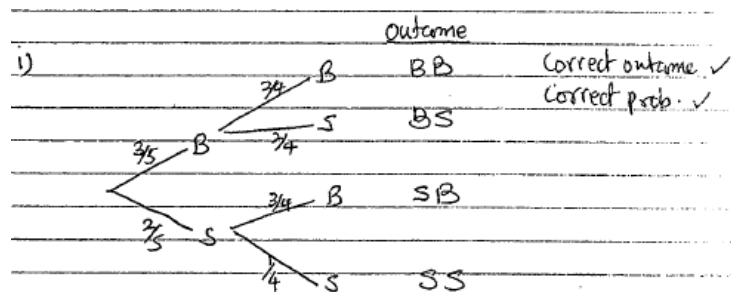
- Draw a probability tree to illustrate the situation.
- What is the probability that Rita selects two cans of baked beans?
- What is the probability that she selects exactly one can of baked beans?
- Rita opens one can and discovers that it is spaghetti. What is the probability that the other can is baked beans?

2

1

2

2



i) $P(\text{select 2B}) = \frac{3}{5} \times \frac{2}{4} = \frac{3}{10}$ ✓ (iv) $\frac{3}{4}$

ii) $P(\text{exactly 1B}) = P(BS) + P(SB)$ ✓
 $= \frac{3}{10} + \frac{3}{10}$
 $= \frac{2}{5}$ ✓

Q14

BAULKHAM

There is one red and three green jelly beans in a jar. One jellybean is selected at random, eaten, and then a second jellybean is selected at random and is also eaten. Find the probability:

- The two jellybeans eaten are both green.
- The red jellybean is the second one eaten.

1

1

i) $P(G, G) = \frac{3}{4} \times \frac{2}{3} = \frac{1}{2}$ ✓
 ii) $P(G, R) = \frac{3}{4} \times \frac{1}{3} = \frac{1}{4}$ ✓

Q15

CARINGBAH

The numbers 0, 1, 2, 3 and 4 are written on 5 discs and placed in a bag. One disc is drawn out at random and the number is recorded, then replaced. A second disc is then drawn out and its number is also recorded.

- What is the probability that the product of the numbers on the two discs is a MULTIPLE of 4?
- What is the probability that the product of the numbers on the two discs is AT LEAST 4?

2

1

$P(\text{multiple of 4}) = \frac{8}{25}$

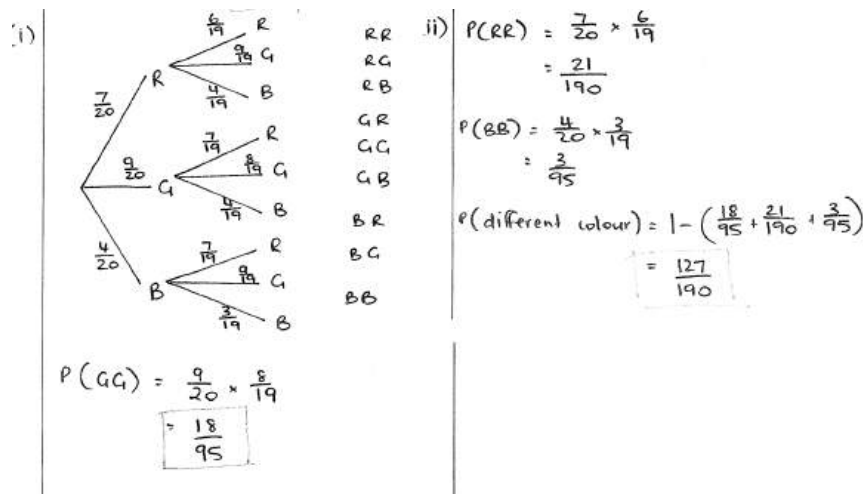
$P(\text{at least 4}) = \frac{11}{25}$

Q16

CARINGBAH

A bag contains 7 red marbles, 9 gold marbles and 4 blue marbles. One marble is drawn from the bag and not replaced, and then the second marble is drawn. Find the probability of choosing:

- two gold marbles. 1
- marbles of different colours. 2



Q17

CSSA

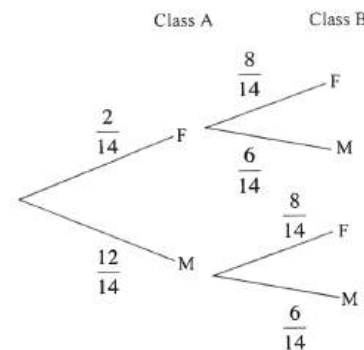
Two students are selected at random to lead a discussion.

One person is chosen from class A, which has 2 females and 12 males, and the other is chosen from class B, which has 8 females and 6 males.

- Find the probability of choosing two females to lead a discussion. 1
- Find the probability of choosing one person of each gender to lead a discussion. 2

(b)(i) (1 mark)

Sample answer:



$$P(FF) = \frac{2}{14} \times \frac{8}{14} = \frac{4}{49}$$

(b)(ii) (2 marks)

Sample answer:

$$P(1 \text{ female, 1 male}) = P(FM) + P(MF)$$

$$= \frac{2}{14} \times \frac{6}{14} + \frac{12}{14} \times \frac{8}{14}$$

$$= \frac{3}{49} + \frac{24}{49} = \frac{27}{49}$$

Q18

CSSA

Fifty tickets are sold in a raffle. There are three prizes. Katherina buys 6 tickets. Which expression gives the probability that Katherina wins none of the three prizes?

- $\frac{6}{50} \times \frac{5}{49} \times \frac{4}{48}$
- $\frac{6}{50} \times \frac{5}{50} \times \frac{4}{50}$
- $\frac{44}{50} \times \frac{43}{49} \times \frac{42}{48}$
- $\frac{44}{50} \times \frac{43}{50} \times \frac{42}{50}$

Q19

CSSA

Xavier is playing a variation of Chess called Makruk on-line. Each game is graded and Xavier begins with a 0.6 probability of winning and a 0.3 probability of losing.

At the end of each game, players receive points.

If Xavier wins he receives 5 points. If he loses he receives 2 points and a draw will result in Xavier receiving 3 points.

(i) What is the probability that Xavier's first game ends in a draw? **1**

In the next game, after grading has occurred, Xavier now has a 0.4 probability of winning and a 0.4 probability of losing.

Find the probability that after two games:

(ii) Xavier receives ten points. **2**

(iii) Xavier receives five points or less. **2**

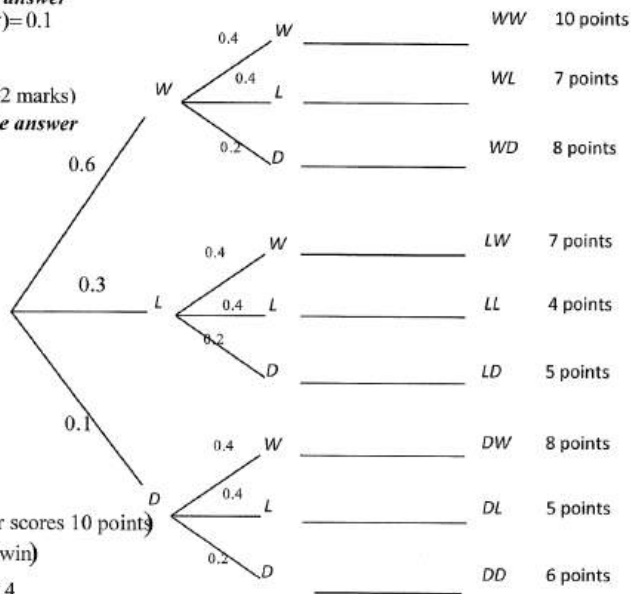
(c) (i) (1 mark)

Sample answer

$P(\text{draw}) = 0.1$

(c) (ii) (2 marks)

Sample answer



$P(\text{Xavier scores 10 points})$

$= P(\text{win/win})$

$= 0.6 \times 0.4$

$= 0.24$

(c) (iii) (2 marks)

Sample answer

$P(\text{Xavier scores 5 points or less}) = P(\text{lose/lose}) + P(\text{lose/draw}) + P(\text{draw/lose})$

$= 0.3 \times 0.4 + 0.3 \times 0.2 + 0.1 \times 0.4$

$= 0.22$

Q20

CSSA

A raffle consists of twenty tickets in which there are two prizes. Darren buys five tickets. First prize is two movie vouchers and second prize is one movie voucher.

The probability that Darren wins at least one movie voucher is

(A) $\frac{5}{20}$

(C) $\frac{7}{16}$

(B) $\frac{27}{76}$

(D) $\frac{17}{38}$

Q21

CSSA

The probability that Matt gets a concert booking with his band on any given weekend is 0.65. What is the probability that over two consecutive weekends he will get at least one concert booking? **2**

$1 - P(\text{no booking, no booking})$

$= 1 - (0.35)^2$

$= 0.8775$

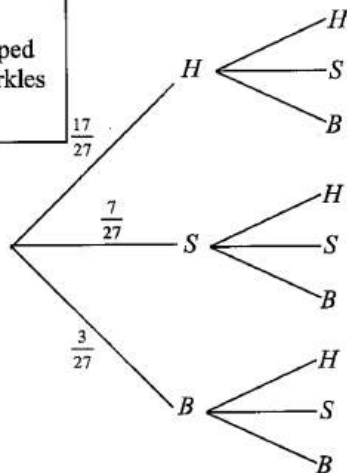
Q22

CSSA

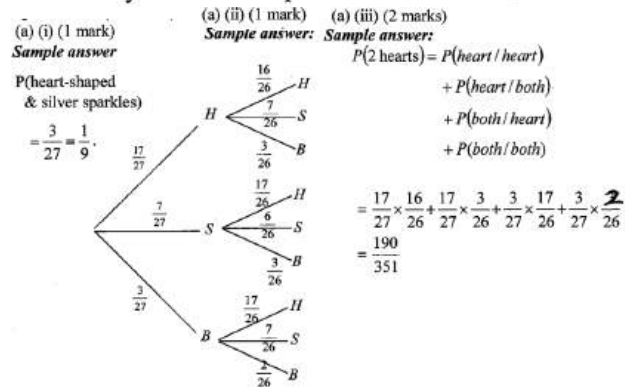
A jar contains 27 buttons. Twenty of the buttons are heart-shaped and ten of the buttons have silver sparkles painted on them. Each button has least one of these two characteristics.

- (i) Explain why the probability that a button chosen at random which is heart-shaped and has silver sparkles is $\frac{1}{9}$.
- (ii) Two buttons are selected from the jar at random without replacement. Copy and complete the tree diagram below indicating the probabilities on each branch.

KEY
H = heart-shaped
S = silver sparkles
B = both



- (iii) What is the probability that when two buttons are drawn that exactly two heart-shaped buttons are selected?



Q23

FORT ST

A factory assembles torches. Each torch requires one battery and one bulb. It is known that 6% of all batteries and 4% of all bulbs are defective.

Find the probability that, in a torch selected at random, both the battery and the bulb are NOT defective. Give your answer in exact form.

6% batteries 4% bulbs $\rightarrow$ defective
94% batteries 96% bulbs $\rightarrow$ good

$$\begin{aligned} P(\text{Both battery and bulb good}) &= 94\% \times 96\% \\ &= \frac{9024}{10000} \\ &= \frac{564}{625} \end{aligned}$$

Q24

FORT ST

Paula has 10 playing cards in her hand, consisting of 5 different pairs of cards. She shuffles the cards and places two of them face up on the table.

- i. Find the probability that these two cards are not a pair
- ii. A third card is placed on the table. Find the probability that this card forms a pair with one of the two cards already on the table.

- i.
 $10 \times 9 = 90$ (sample space)
 $10 \times 8 = 80$ (for 2nd card not matching)
 $\therefore P(\text{not a pair}) = \frac{80}{90} = \frac{8}{9}$
- ii.
 $10 \times 8 \times 2 = 160$
 $10 \times 9 \times 8 = 720$
 $\therefore P(3\text{rd a pair}) = \frac{160}{720} = \frac{2}{9}$

Q25

FORT ST

There are 5 red marbles and 4 blue marbles in a bag. Bill and Ben are playing a game in which they take turns drawing a marble from the bag and then replacing it.

To win the game, Ben must draw a red marble and for Bill to win he must draw a blue marble. They continue taking turns until there is a winner. Ben goes first.

- Find the probability that Ben wins on his first draw. 1
- Find the probability that Ben wins in three or less of his turns. 2
- Find the probability that Ben wins the game. 2

5R, 4B

Ben Wins in 3 or less turns

i) $\frac{5}{9}$ ✓

ii) $\frac{5}{9} + \frac{4}{9} \cdot \frac{5}{9} \cdot \frac{5}{9} + \frac{4}{9} \cdot \frac{5}{9} \cdot \frac{4}{9} \cdot \frac{5}{9} \cdot \frac{5}{9}$ ✓

$= \frac{5}{9} + (\frac{5}{9})^2 \cdot \frac{4}{9} + (\frac{5}{9})^3 \cdot (\frac{4}{9})^2$ ✓

$\approx \frac{505}{729} + \frac{2000}{59049} \approx 0.726599942$

Many student could not generate the seq.

iii) Ben Wins $S_{\infty} = \frac{\frac{5}{9}}{1 - \frac{20}{81}} = \frac{5}{9} \cdot \frac{81}{61} = \frac{45}{61}$ ✓

$P(\text{Ben Wins}) = \frac{45}{61} \approx 0.737704918$ ✓

many student did not realise the Seq

Q26

GIRAWEEEN

A standard pack of 52 cards consists of four suits (Diamonds, Hearts, Clubs and Spades) with 13 cards in each suit.

- One card is drawn from the pack and kept on the table. A second card is drawn and placed beside it on the table. 1
What is the probability that the second card is from a different suit to the first?
- The two cards are replaced and the pack shuffled. Four cards are chosen from the pack and placed side by side on the table. 2
What is the probability that these four cards are all from different suits?

i) After first card is drawn, 51 cards remain of which 3x13 are of a different suit

$P(\text{2nd card different suit}) = \frac{39}{51} = \frac{13}{17}$ (1m)

5)(a)(ii)

$P(\text{all 4 different suits}) = P(\text{card 1 any suit}) \times P(\text{card 2 any of 3 suits}) \times P(\text{card 3 any of 2 suits}) \times P(\text{card 4 last remaining suit})$

$= (\frac{52}{52}) \times (\frac{39}{51}) \times (\frac{26}{50}) \times (\frac{13}{49})$

$= \frac{685464}{6497400}$

$= \frac{2197}{20825}$

$= 0.1054981993...$ (2m)

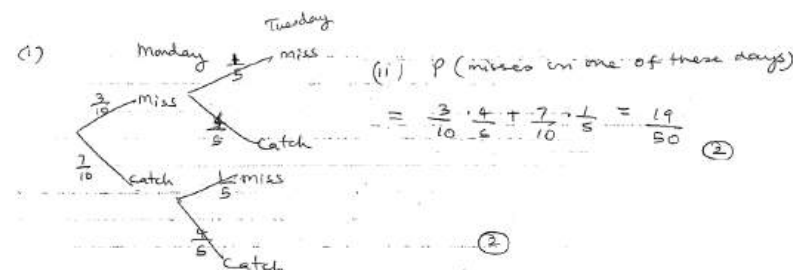
≈ 0.105 (3 decimal places)

Q27

GIRAWEEEN

Vincent travels to school every day by train. He has found that the probability he misses his train on Mondays is 30% while on Tuesdays it is 20%.

- Draw a probability tree diagram for Monday and Tuesday. 2
- Find the probability that Vincent misses the train on ONE of these days. 2



Q28

GIRAWEEEN

A bag contains six discs. Two of the discs have the number 0 on them and the other four discs have the number 1 on them. Three discs are drawn at random without replacement.

- Find the probability that all of the three discs drawn have the number 1 on them. [2]
- Find the probability that the product of the numbers on the three discs drawn is 0. [1]

$$(i) p = \frac{4}{6} \times \frac{3}{5} \times \frac{2}{4} = \frac{1}{5}$$

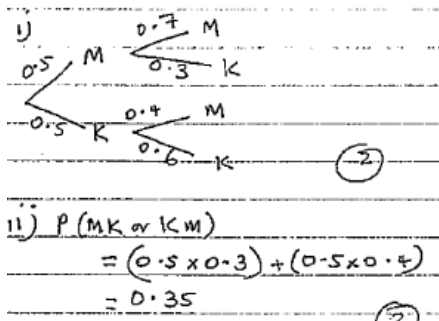
$$(ii) p(\text{at least one } 0) = 1 - p(\text{all } 1's) \\ = 1 - \frac{1}{5} = \frac{4}{5}$$

Q29

GIRAWEEEN

Monique and Kaavya are playing in a tennis tournament and will play 2 sets. Each has an equal chance of winning the first set. If Monique wins the first set, the probability of her winning the second set is 0.7. If Monique loses the first set, the probability of her winning the second set is 0.4.

- Draw a probability tree to represent this information. 2
- Find the probability that Monique wins exactly one set. 2

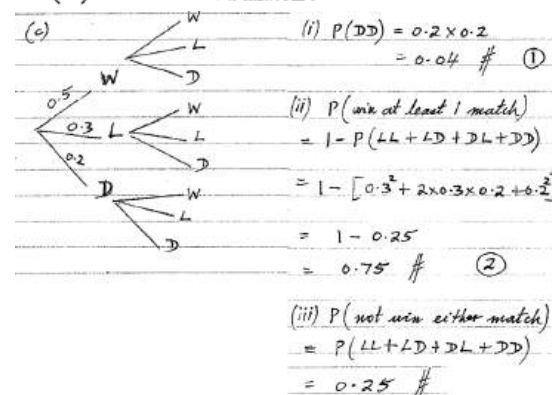


Q30

GIRAWEEEN

A certain soccer team has a probability of 0.5 of winning a match and a probability of 0.2 of drawing the match. If the team plays two matches, find the probability that it will :

- draw both matches. 1
- win at least one match. 2
- not win either match. 2



Q31

GOSFORD

A total of 300 tickets are sold in a raffle which has 3 prizes. There are 100 red, 100 green and 100 blue tickets. At the drawing of the raffle, winning tickets are NOT replaced before the next draw.

- What is the probability that each of the 3 winning tickets is red? 1
- What is the probability that at least one of the winning tickets is not red? 1
- What is the probability that there is one winning ticket of each colour? 2

$$i) \frac{100}{300} \times \frac{99}{299} \times \frac{98}{298} = \frac{1617}{44551}$$

$$ii) 1 - \frac{1617}{44551} = \frac{42934}{44551} = 0.964$$

$$iii) \frac{100}{300} \times \frac{100}{299} \times \frac{100}{298} \times 6 = \frac{10000}{44551} = 0.224$$

Q32

GOSFORD

An engine under development contains three components that are prone to failure. The probabilities that each of the components will fail are $\frac{1}{10}$, $\frac{1}{5}$ and $\frac{1}{4}$ respectively. What is the probability that at least one part will fail?

(A) $\frac{23}{50}$

(B) $\frac{1}{200}$

(C) $\frac{3}{100}$

(D) $\frac{11}{20}$

$$\begin{aligned} P(\geq 1 \text{ fail}) &= 1 - P(0 \text{ fail}) \\ &= 1 - \left(\frac{9}{10} \times \frac{4}{5} \times \frac{3}{4} \right) \\ &= 1 - \frac{27}{50} \\ &= \frac{23}{50} \quad (A) \end{aligned}$$

Q33

GOSFORD

Four cards, numbered 3, 4, 5 and 6 are used in a game.

3

4

5

6

The four cards are placed face down and each player pays \$1 to take a turn to draw two cards, one at a time without replacement.

The two cards make a two digit number, with the first card drawn being the first digit of their number.

If the cards form a number over 60, the player receives \$3 back (i.e. they win \$2), otherwise, they receive nothing back.

The four cards are then shuffled and replaced for the next turn.

- Use a diagram to show all of the possible 2-digit numbers that could be drawn. 1
- What is the probability that a player will win on their first turn? 2
- Calculate the probability that a player who brings \$5 to play will have \$3 left after 5 turns? 2

(i)	2 nd card.	(iii)	Start with \$5.
	3 4 5 6		5 turns cost 5 x \$1
1 st card	3 X 34 35 36		\$0 remaining
	4 43 X 45 46		1 win = +\$3 : 4 losses
	5 53 54 X 56		
	6 63 64 65 X		

$$\begin{aligned} P(\text{win, 4 losses}) &= \frac{1}{4} \times \left(\frac{3}{4} \right)^4 \times 5 \\ &= \frac{405}{1024} \end{aligned}$$

(ii) $P(w) = \frac{1}{4}$

Q34

HUNTERS
HILL

A card is drawn at random from a standard deck of playing cards. What is the probability that the card drawn is a Diamond OR a Queen?

(A) $\frac{13}{52}$
(B) $\frac{4}{52}$

(C) $\frac{17}{52}$
(D) $\frac{16}{52}$

Q35

HUNTERS
HILL

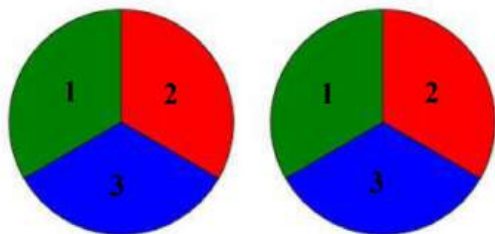
An archer fires three arrows at a target. The probability of any single arrow hitting the target is $\frac{4}{5}$. Find the probability that:

- The first arrow hits and the next two miss. 1
- The archer hits the target exactly once. 1
- The archer hits the target at least once. 1

$$\begin{aligned} \text{i)} \quad P(H, M, M) &= \frac{4}{5} \cdot \frac{1}{5} \cdot \frac{1}{5} = \frac{4}{125} \\ \text{ii)} \quad P(\text{one H}) &= \frac{4}{5} \times 3 = \frac{12}{125} \\ \text{iii)} \quad P(\text{at least one H}) &= 1 - P(3 M) \\ &= 1 - \frac{1}{5} \cdot \frac{1}{5} \cdot \frac{1}{5} \\ &= \frac{124}{125} \end{aligned}$$

Q36

HURLSTONE



Two identical spinners, containing the values 1, 2, and 3 are spun and the results on each are added together. What is the probability that the resulting sum is an even number?

A: $\frac{1}{3}$

B: $\frac{5}{9}$

C: $\frac{2}{3}$

D: $\frac{7}{9}$

Q37

HURLSTONE

Two-digit numbers are formed from the digits 2, 3, 4, 5, 6. Repetition of digits is allowed. A two-digit number is then selected at random. What is the probability the number is a multiple of 3?

2

There are 8 multiples of 3: 24, 33, 36, 42, 45, 54, 63, 66

$$P(\text{multiple of 3}) = \frac{8}{25}$$

Q38

HURLSTONE

A game is played in which two coloured dice are thrown once. The six faces of the red die are numbered 1, 3, 5, 7, 9 and 11. The six faces of the white die are numbered 2, 4, 6, 8, 10 and 12. The player wins if the number on the red die is larger than the number on the white die.

- Show that the probability of the player winning a game is $\frac{5}{12}$. 1
- What is the probability that the player wins exactly once in two successive games? 2
- What is the probability that the player wins at least once in two successive games? 1

(e) (i) $P(\text{win}) = \frac{0+1+2+3+4+5}{6 \times 6} = \frac{15}{36} = \frac{5}{12}$

(ii)
$$P(\text{Exactly one from two}) = P(WL) + P(LW)$$

$$= \frac{5}{12} \times \frac{7}{12} + \frac{7}{12} \times \frac{5}{12}$$

$$= \frac{70}{144} \text{ or } \frac{35}{72}$$

(iii)
$$P(\text{At least 1 win from 5}) = 1 - P(\text{No wins})$$

$$= 1 - \left(\frac{7}{12}\right)^2 = \frac{95}{144}$$

Q39

HURLSTONE

There are two prizes in a raffle in which 50 tickets are sold. The first prize is obtained by drawing a ticket at random and this ticket is not replaced for the draw of the second prize. Deepti buys two tickets in the raffle. What is the probability that she does not win a prize?

(A) $\frac{576}{625}$

(B) $\frac{47}{50}$

(C) $\frac{1128}{1225}$

(D) $\frac{1152}{1225}$



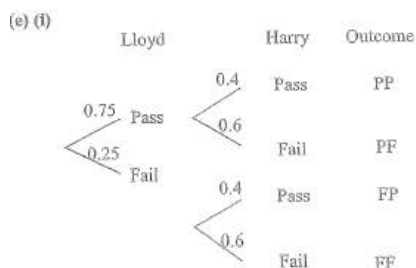
Q40

HURLSTONE

There is a 75% chance that Lloyd will pass his driving test and a 40% chance that

Harry will pass his.

- (i) Draw a probability tree to represent this information, including the list of possible outcomes. 2
- (ii) Find the probability that only one of Lloyd and Harry pass the test. 1



$$(iii) P(\text{Only 1 passes}) = P(PF) + P(FP)$$

$$= 0.75 \times 0.6 + 0.25 \times 0.4$$

$$= 0.55 \text{ or } 55\%$$

Q41

HURLSTONE

The chance of a fisherman catching a legal length fish is 4 in 5. If three fish are caught at random, what is the probability that exactly one is of legal length?

(A) $\frac{4}{125}$

(B) $\frac{12}{125}$

(C) $\frac{16}{125}$

(D) $\frac{48}{125}$

Q42

HURLSTONE

Each week Lucinda and Terry take part in a raffle at their respective workplaces.

The probability that Lucinda wins a prize in her raffle is $\frac{1}{9}$.

The probability that Terry wins a prize in his raffle is $\frac{1}{16}$.

What is the probability that, during the next three weeks, at least one of them wins a prize? 2

$$P(\text{atleast one}) = 1 - P(\text{none win})$$

$$= 1 - \left(\frac{8}{9}\right)^3 \times \left(\frac{15}{16}\right)^3$$

$$= \frac{91}{216}$$

Q43

HURLSTONE

In a certain city, the probability that the pollution level will be high is 0.3. If the pollution is monitored for 3 successive days, find the probability that the pollution levels will be:

- (i) High on each of the 3 days 1
- (ii) High on exactly 2 days 1

(b)
 $P(\text{high}) = 0.3$

(i)
 $P(H, H, H) = 0.3^3$
 $= 0.027$

(ii)
 $P(H, H, _) + P(H, _, H) + P(_, H, H)$
 $= 3 \times (0.3^2 \times 0.7)$
 $= 0.189$

Q44

INDEPENDENT

Bag A contains 4 blue balls and 3 red balls. Bag B contains 2 blue balls and 3 red balls. Pierre chooses one ball at random from the 7 balls in bag A and puts it in bag B. He then chooses one ball at random from the 6 balls in bag B. Find the probability he chooses a red ball from bag B.

$$P(\text{red from A then a red from B}) = \frac{3}{7} \times \frac{4}{6}$$

$$P(\text{blue from A then a red from B}) = \frac{4}{7} \times \frac{3}{6}$$

$$\therefore P(\text{a red from B}) = \frac{3}{7} \times \frac{4}{6} + \frac{4}{7} \times \frac{3}{6} = \frac{4}{7}$$

Q45

INDEPENDENT

Blaise rolls a fair die repeatedly until he first rolls a 5 or a 6.

- Find as a fraction in its simplest terms the probability that Blaise rolls the die exactly 8 times. 2
- Find as a fraction in its simplest terms the probability that Blaise rolls the die fewer than 8 times. 3

i. On each roll, $P(\text{rolls 5 or 6}) = \frac{2}{6} = \frac{1}{3}$. Hence $P(\text{does not roll 5 or 6}) = \frac{2}{3}$.

$$P(\text{Blaise rolls exactly 8 times}) = \left(\frac{2}{3}\right)^7 \left(\frac{1}{3}\right) = \frac{128}{6561}$$

ii. $P(\text{fewer than 8 rolls}) = 1 - P(\text{no 5's or 6's on first 7 rolls}) = 1 - \left(\frac{2}{3}\right)^7 = \frac{2059}{2187}$

Q46

INDEPENDENT

There are 2 piles of playing cards. One pile contains 7 red and 4 black cards while the other pile contains 3 red and 4 black cards.

Felicity randomly chooses one card from each pile.

- What is the probability that both cards are black? 1
- What is the probability that at least one card is red? 1
- What is the probability that both cards are different colours? 2

c)(i) Outcome Assessed: HS

c)(ii) Outcome Assessed: HS

c)(iii) Outcome Assessed: HS

• Correct solution

• Correct solution

Answer:

$$\text{Pile 1: } P(B) = \frac{4}{11}, \quad P(R) = \frac{7}{11}$$

$$\text{Pile 2: } P(B) = \frac{4}{7}, \quad P(R) = \frac{3}{7}$$

$$P(BB) = P(B) \times P(B)$$

$$= \frac{4}{11} \times \frac{4}{7} = \frac{16}{77}$$

Answer:

$$P(\text{at least 1 red}) = 1 - P(BB)$$

$$= 1 - \frac{16}{77}$$

$$= \frac{61}{77}$$

Answer:

$$P(\text{both cards different}) = P(RB) + P(BR)$$

$$= \frac{7}{11} \times \frac{4}{7} + \frac{4}{11} \times \frac{3}{7}$$

$$= \frac{28}{77} + \frac{12}{77}$$

$$= \frac{40}{77}$$

Q47

INDEPENDENT

Paula has 10 playing cards in her hand, consisting of 5 different pairs of cards. She shuffles the cards and places two of them face up on the table.

- Find the probability that these two cards are not a pair. 1
- A third card is placed on the table. Find the probability that this card forms a pair with one of the two cards already on the table. 1

$$\begin{array}{l|l} \text{(i)} & P(\text{pair}) = \frac{8}{9} \\ \text{(ii)} & P(\text{pair}) = \frac{2}{8} \\ & = \frac{1}{4} \end{array}$$

Q48

INDEPENDENT

During July the probability that it rains on any day is $\frac{1}{3}$.

Find the probability that during a 7 day week

- it rains on the 1st and 5th day.
- it rains only on the first 3 days.
- there is at least one rainy day.

1
1
2

b)(i)	$P(E) = \frac{1}{3} \times \frac{1}{3}$ $= \frac{1}{9}$	b)(ii)	$P(E) = \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{1}{3} \cdot \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{2}{3} \cdot \frac{2}{3}$ $= \frac{16}{2187}$
b)(iii)	$P(\text{at least one rainy day}) = 1 - P(\text{all dry days})$ $= 1 - \left(\frac{2}{3}\right)^7$ $= 1 - \frac{128}{2187}$ $= \frac{2059}{2187}$		

Q49

INDEPENDENT

Fiona is planting exotic flowers in her garden. The probability that an individual flower survives is 0.35. Fiona plants 5 flowers.

What is the probability that at least one flower survives?

Sample Answer:-

$$P(\text{Die}) = 1 - 0.35$$

$$= 0.65$$

$$P(\text{at least one survives}) = 1 - P(\text{all die})$$

$$= 1 - (0.65)^5$$

$$= 0.8839709375$$

Q50

A pack of 52 playing cards is divided into 4 equal suits: diamonds, hearts, clubs and spades.

2

NEAP

What is the probability of picking two cards (without replacement) from the same suit?

$$P(\text{picking two cards from same suit}) = \frac{13}{52} \times \frac{12}{51}$$

$$= \frac{1}{17}$$

$$P(\text{the two cards came from one of the four suits}) = 4 \times \frac{1}{17}$$

$$= \frac{4}{17}$$

Q51

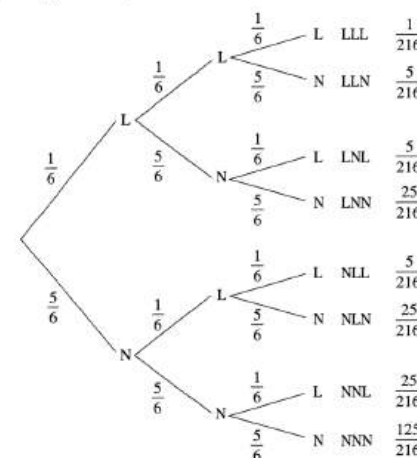
Jim finds it difficult to get out of bed each day. The probability that he will be late for work on any particular day is $\frac{1}{6}$. His employer checks if Jim is on time on three successive days.

NEAP

- Draw a tree diagram displaying all the possible outcomes and probabilities.
- Find the probability that Jim is on time on all three days.
- Find the probability that Jim is on time on two days and late on one day.

2
1
1

(c) (i) L = late, N = not late



$$(ii) P(\text{not late}) = \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6}$$

$$= \frac{125}{216}$$

$$(iii) P(\text{late only one day}) = P(LNN) + P(NLN) + P(NNL)$$

$$= \frac{25}{216} + \frac{25}{216} + \frac{25}{216}$$

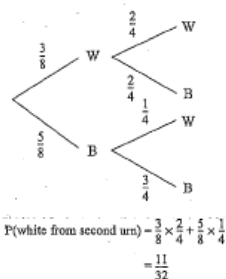
$$= \frac{75}{216}$$

Q52

An urn contains 3 white and 5 black balls. A second urn contains 1 white and 2 black balls. A ball is drawn at random from the first urn and placed in the second urn.

NEAP

If a ball is then drawn from the second urn, what is the probability that the ball is white?



Q53

NEAP

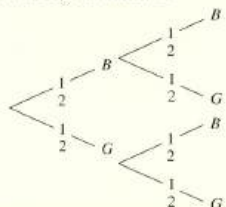
The probability that any pregnant Australian woman will have twins is $\frac{1}{60}$.

- What is the probability that a pregnant Australian woman will not have twins?
- Maria's three daughters are pregnant. One of Maria's daughters is pregnant with twin boys and the other two daughters are expecting single babies of unknown sex.

What is the probability that at least one of Maria's expected four grandchildren will be female? Assume that males and females are equally likely.

(i) $\frac{59}{60}$

(ii) The twin boys aren't relevant.



$1 - \text{probability that both (unknown sex) babies are boys}$
 $= 1 - \left(\frac{1}{2}\right)^2 = \frac{3}{4}$

Q54

SBHS

p is an integer chosen at random from the set $\{5, 7, 9, 11\}$

q is an integer chosen at random from the set $\{2, 6, 10, 14, 18\}$

What is the probability that $p + q = 23$?

- 0.1
- 0.2
- 0.3
- 0.4

The only options are (5, 18) and (9, 14).

There are $4 \times 5 = 20$ options

$$P(p + q = 23) = \frac{2}{20} = 0.1$$

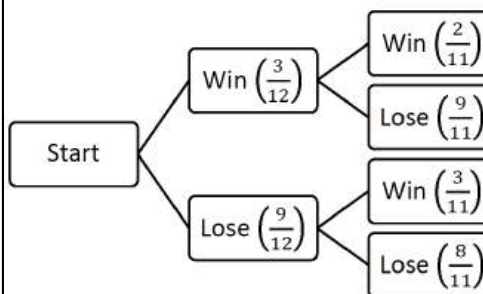
Q55

SBHS

In a lucky dip, there are twelve identical envelopes of which only three contain prizes.

- Show that if one were to purchase two envelopes the probability of not getting a prize would be $\frac{6}{11}$.

- What is the probability of getting at least one prize in this case?



$$\begin{aligned} P(\text{No prize at all}) &= P(\text{No prize in first}) \times P(\text{No prize in second}) \\ &= \frac{9}{12} \times \frac{8}{11} \\ &= \frac{6}{11} \end{aligned}$$

Part C, Subsection II:

$$\begin{aligned} P(\text{At least one prize}) &= 1 - P(\text{No prize at all}) \\ &= 1 - \frac{6}{11} \\ &= \frac{5}{11} \end{aligned}$$

Q56

SBHS

Fifty tickets are sold in a raffle. There are two prizes. Michelle buys 5 tickets. The probability that she does not win either prize is given by

(A) $1 - \frac{5}{50} \times \frac{4}{49}$

(C) $\frac{45}{50} \times \frac{44}{50}$

(B) $\frac{45}{50} + \frac{44}{49}$

(D) $\frac{45}{50} \times \frac{44}{49}$

Q57

SBHS

Two dice are biased so that the probability of throwing a six, face up, is $\frac{3}{8}$ and the probability the probability of each of the other five numbers is $\frac{1}{8}$. Find the probability of:

- (i) rolling a double six, face up; 1
- (ii) only one six appearing face up when the two dice are rolled. 2

(a) $\left(\frac{3}{8}\right)^2 = \frac{9}{64}$ [1]

(i)

(ii) $\left(\frac{3}{8}\right)\left(\frac{5}{8}\right) \times 2$

$= \frac{30}{64}$

$= \frac{15}{32}$ [2]

Q58

SBHS

One date is taken at random from those in May and another from those in June. (2)

What is the probability that both are the 13th of the month? (Leave your answer as a fraction)

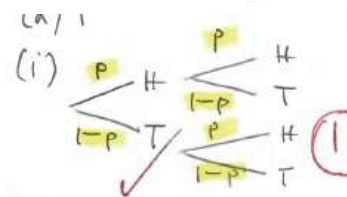
$\frac{1}{31} \times \frac{1}{30} = \frac{1}{930}$

Q59

SBHS

The probability that a biased coin lands heads up is p . I have a pair of identical biased coins. The coins show heads more often than normal coins show heads.

- (i) Draw a tree diagram to show the possible outcomes when I toss the pair of biased coins together. Write the probabilities on each branch of the tree. (1)
- (ii) When I toss the coins, 25% of the time they land showing a head and a tail. Determine the probability that, on the next toss of the pair of coins, they will land with at least one of the coins showing a head. Give your answer correct to 3 decimal places. (3)



(ii) $2p - 2p^2 = \frac{1}{4}$
 $8p^2 - 8p + 1 = 0$
 $p = \frac{8 \pm 4\sqrt{2}}{16}$

$\left(\frac{2+\sqrt{2}}{4}\right) \left(\frac{2-\sqrt{2}}{4}\right) \times 2$
 0.8535×0.1465

$P(\text{At least 1 head}) = 1 - P(TT)$

$p = 0.8535$
 $1 - p^2 = 0.9199$
 Note: $p > \frac{1}{2}$ as heads is more likely



Probability

Part A: tree diagram



Q60

SGHS

A raffle consists of twenty tickets in which there are two prizes. Mike buys five tickets. First prize is two movie vouchers and second prize is one movie voucher. The probability that Mike wins at least one movie voucher is

(A) $\frac{17}{38}$

(C) $\frac{7}{16}$

(B) $\frac{27}{76}$

(D) $\frac{5}{20}$

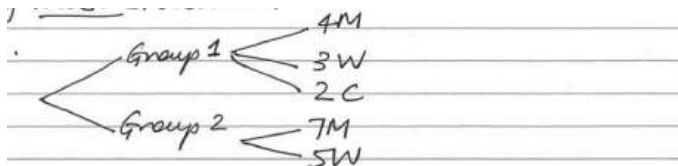
Q61

SBHS

There are two groups of people at a party and Minh is blind-folded. In the first group there are 4 men, 3 women and 2 children. In the second group there are 7 men and 5 women.

Minh is spun around and asked to select one person at random.

- Find the probability that Minh approaches the first group and then selects a woman. 1
- Find the probability that a woman from either group is selected. 2



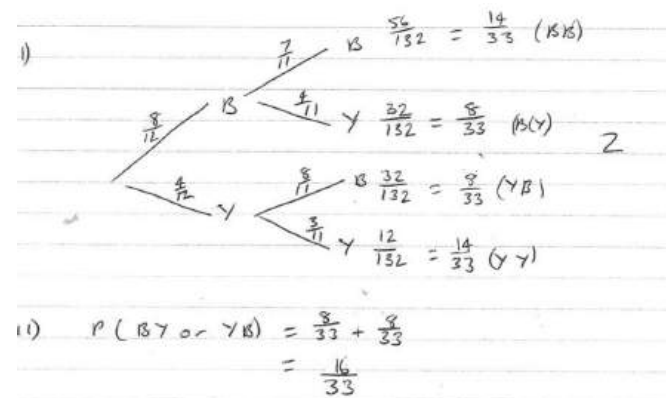
i) $P(G1, W)$	ii) $P(G1, W) \text{ or } P(G2, W)$
$= \frac{1}{2} \times \frac{3}{9}$	$= \frac{1}{2} \times \frac{3}{9} + \frac{1}{2} \times \frac{5}{12}$
$= \frac{3}{18}$	$= \frac{1}{6} + \frac{5}{24}$
$= \frac{1}{6}$	$= \frac{3}{6}$

Q62

SGHS

A bag contains 8 blue marbles and 4 yellow marbles. Two marbles are selected at random without replacement.

- Draw a tree diagram to show all possible outcomes. Include the probability on each branch. [2]
- What is the probability that the two marbles are of different colours? [1]



Q63

SGHS

Liam, Harry and Zach work independently on a problem. If the respective probabilities that they will solve it are $\frac{3}{4}$, $\frac{1}{2}$ and $\frac{2}{5}$, find the probability that the problem will be solved.

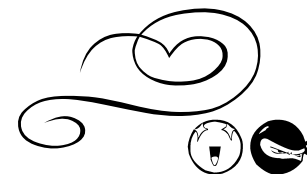
Probability at least one solves problem
 $= 1 - \text{Probability that no one solves problem}$
 $P = 1 - \left(\frac{1}{4} \times \frac{1}{2} \times \frac{3}{5} \right)$
 $= 1 - \frac{3}{40}$
 $P = \frac{37}{40}$ (2 marks)

→ This question was done well overall. Some over complicated



Probability

Part B: Lattice



Q 1

ACE

A game is played in which two coloured dice are thrown once. The six faces of the red die are numbered 3, 5, 7, 8, 9 and 11. The six faces of the white die are numbered 1, 2, 4, 6, 10 and 12. The player wins if the number on the white die is larger than the number on the red die.

- What is the probability of the player winning a game? 1
- What is the probability that the player wins once in two successive games? 2
- What is the probability that the player wins at least once in two successive games? 1

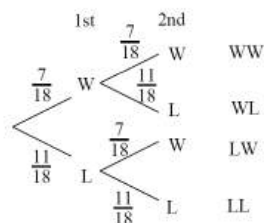
Red die	White die					
	1	2	4	6	10	12
3						
5						
7						
8						
9						
11						

White die is larger than the red die on 14 occasions.

$$\begin{aligned} \text{(a)} \quad P(W) &= \frac{14}{36} \\ \text{(i)} \quad &= \frac{7}{18} \end{aligned}$$

15(a)

(ii)



$$\begin{aligned} P(WL \text{ or } LW) &= \frac{11}{18} \times \frac{7}{18} + \frac{7}{18} \times \frac{11}{18} \\ &= \frac{77}{162} \end{aligned}$$

15(a)

(iii)

Probability of at least one win is every outcome except losing both games.

$$\begin{aligned} 1 - P(LL) &= 1 - \frac{11}{18} \times \frac{11}{18} \\ &= \frac{203}{324} \end{aligned}$$

Q 2

FORT ST

Two-digit numbers are formed from the digits 2, 3, 4, 6 with no repetition of digits allowed. A two-digit number is then selected at random. What is the probability that the number is even?

- $\frac{1}{12}$
- $\frac{1}{8}$

$$\text{(C)} \quad \frac{3}{4}$$

$$\text{(D)} \quad \frac{5}{12}$$

	2	3	4	6
2	X		✓	✓
3	✓	X	✓	✓
4	✓		X	✓
6	✓		✓	X

Total number of outcomes = 12
Favourable outcomes = 9

$$\begin{aligned} P(\text{Even}) &= \frac{9}{12} \\ &= \frac{3}{4} \end{aligned}$$

Q 3

FORT ST

Pat and Chris each threw a die

- Find the probability that they throw the same number. 1
- Find the probability that the number thrown by Chris is greater than the number thrown by Pat. 1

(c) (i) Probability

$$\begin{aligned} &= P[(1,1) \text{ or } (2,2) \text{ or } (3,3) \text{ or } (4,4) \text{ or } (5,5) \text{ or } (6,6)] \\ &= \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} + \frac{1}{36} \\ &= \frac{6}{36} = \frac{1}{6} \end{aligned}$$

Probability (Chris's throw same as Pat's) = $\frac{1}{6}$

(ii) Let Pat be first, Chris be second relevant pairs are

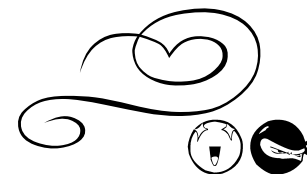
(1,2) (1,3) (1,4) (1,5) (1,6)
(2,3) (2,4) (2,5) (2,6)
(3,4) (3,5) (3,6)
(4,5) (4,6)
(5,6)

$$P(\text{Chris's no.} > \text{Pat's no.}) = \frac{15}{36} = \frac{5}{12}$$



Probability

Part B: Lattice



2unit

selective

Q5

Mayomi and Christina each throw a die.

GIRAEEN

- Find the probability that they throw the same number.
- Find the probability that the number thrown by Christina is greater than the number thrown by Mayomi.

12) P (throw same number)

$$= P(1,1) \text{ or } P(2,2) \text{ or } P(3,3) \text{ or } P(4,4) \\ \text{or } P(5,5) \text{ or } P(6,6)$$

$$= 6 \times \frac{1}{36}$$

$$= \frac{6}{36}$$

$$= \frac{1}{6} \quad (2m)$$

12) F(ii) Let Mayomi be first number, then Christina throw second number.
Possible outcomes:

(1,1) (1,2) (1,3) (1,4) (1,5) (1,6)
(2,1) (2,2) (2,3) (2,4) (2,5) (2,6)
(3,1) (3,2) (3,3) (3,4) (3,5) (3,6)
(4,1) (4,2) (4,3) (4,4) (4,5) (4,6)
(5,1) (5,2) (5,3) (5,4) (5,5) (5,6)
(6,1) (6,2) (6,3) (6,4) (6,5) (6,6)

15 possible outcomes where Christina throws number greater than Mayomi

$$\therefore P(\text{Christina's number} > \text{Mayomi's number})$$

$$= \frac{15}{36}$$

$$= \frac{5}{12} \quad (2m)$$

12) F(ii) ALTERNATIVE METHOD

from F(i), $P(\text{throw same number}) = \frac{1}{6}$
 $\therefore P(\text{do NOT throw same number}) = \frac{5}{6}$
 of this, $\frac{1}{2} \times \frac{5}{6} = P(\text{Christina} > \text{Mayomi})$
 $= P(\text{Mayomi} > \text{Christina})$
 $\therefore P(\text{Christina's number} > \text{Mayomi's number})$
 $= \frac{1}{2} \times \frac{5}{6}$

$$= \frac{5}{12} \quad (2m)$$

Q6

HURLSTONE

A game is played in which two coloured dice are thrown once. The six faces of the red die are numbered 1, 3, 5, 7, 9 and 11. The six faces of the white die are numbered 2, 4, 6, 8, 10 and 12. The player wins if the number on the white die is larger than the number on the red die.

- Show that the probability of the player winning a game is $\frac{7}{12}$. 1
- What is the probability that the player wins exactly once in two successive games? 2
- What is the probability that the player wins at least once in two successive games? 1

(d) (i)

	1	3	5	7	9	11
2	Win	Lose	L	L	L	L
4	W	W	L	L	L	L
6	W	W	W	L	L	L
8	W	W	W	W	L	L
10	W	W	W	W	W	L
12	W	W	W	W	W	W

$$P(\text{Win}) = \frac{21}{36} = \frac{7}{12}$$

$$(ii) P(\text{Exactly one win from two}) = \frac{7}{12} \times \frac{5}{12} + \frac{5}{12} \times \frac{7}{12}$$

$$= \frac{70}{144} \text{ or } \frac{35}{72}$$

$$(iii) P(\text{Win at least once}) = 1 - P(\text{lose both})$$

$$= 1 - \frac{5}{12} \times \frac{5}{12}$$

$$= \frac{119}{144}$$



Probability

Part B: Lattice



Q7

INDEPENDENT

Two ordinary dice are rolled. The "score" is the sum of the numbers on the top faces. What is the probability that the score is 9?

A $\frac{3}{4}$

B $\frac{1}{3}$

C $\frac{1}{4}$

$(6,3)(5,4)(4,5)(3,6)$

D $\frac{1}{9}$ $P(9) = \frac{4}{36}$
 $= \frac{1}{9}$

Q8

INDEPENDENT

Jeff and Jonas each throw a standard die.

(i) Find the probability that they do *not* throw the same number. 1

(ii) Find the probability that the number thrown by Jonas is less than the number thrown by Jeff. 1

	Jo 1	Jo 2	Jo 3	Jo 4	Jo 5	Jo 6
Je 1	1, 1	1, 2	1, 3	1, 4	1, 5	1, 6
Je 2	2, 1	2, 2	2, 3	2, 4	2, 5	2, 6
Je 3	3, 1	3, 2	3, 3	3, 4	3, 5	3, 6
Je 4	4, 1	4, 2	4, 3	4, 4	4, 5	4, 6
Je 5	5, 1	5, 2	5, 3	5, 4	5, 5	5, 6
Je 6	6, 1	6, 2	6, 3	6, 4	6, 5	6, 6

a)(i) $P(\text{different}) = \frac{5}{6}$

a)(ii) $P(E) = \frac{15}{36}$
 $= \frac{5}{12}$

Q9

SBHS

Two fair dice are thrown. If the first die thrown is a 3, what is the probability that the sum of the two dice is greater than 7?

(A) $\frac{1}{18}$

(B) $\frac{1}{3}$

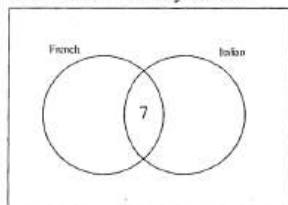
(C) $\frac{1}{2}$

(D) $\frac{1}{12}$

Q 1

BAULKHAM

The Venn Diagram represents a group of 30 students all of whom study either Italian, French or both languages. 7 students study both and 19 students study French.



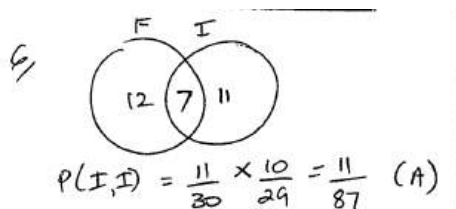
By completing the Venn Diagram or otherwise, find the probability that if 2 people are selected at random that they only study Italian.

(A) $\frac{11}{87}$

(C) $\frac{22}{145}$

(B) $\frac{2}{145}$

(D) $\frac{57}{145}$



Q 2

CSSA

In a class of 30 girls, 13 are dancers and 23 are gymnasts. If 7 girls do both dance and gymnastics, what is the probability that a girl chosen at random does neither dance nor gymnastics?

(A) $\frac{1}{5}$

(B) $\frac{7}{30}$

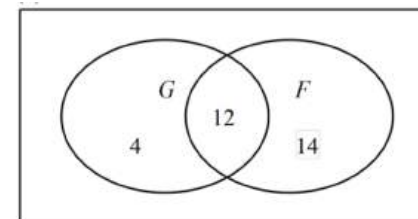
(C) $\frac{1}{30}$

(D) $\frac{8}{15}$

Q3

HORNSBY
GIRLS

A language class has students who study at least one of the languages French and German. Of the class of 30 students, 16 study German and 26 study French. What is the probability that a student selected at random studies both languages?

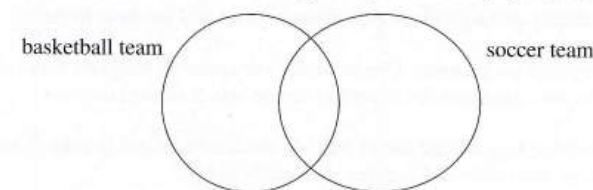


$$P(\text{study both languages}) = \frac{12}{30} = \frac{2}{5}$$

Q4

NEAP

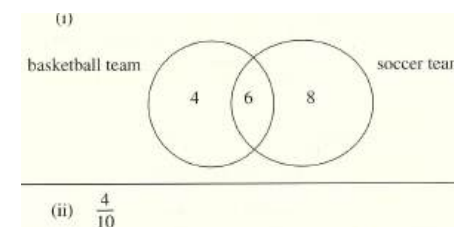
The Wolves are a basketball team and the Hammers are a soccer team. A total of 18 people are members of one or the other or both teams. The soccer team has 14 members and the basketball team has 10 members. The diagram represents the players in the teams.



(i) Copy the diagram and place appropriate numbers in each section. 1

(ii) A player is selected at random from the basketball team.

What is the probability that the player is **not** in the soccer team? 1





Probability

Part D: finding numbers



Q1

BARKER

A horticulturist found the probability that a planted tomato seedling will eventually bear fruit was 0.75. He planted 'n' seedlings.

- What is the probability that no seedlings will bear fruit?
- How many seedlings must be planted to be at least 99% certain that at least one seedling will bear fruit?

$$1) P(\text{no fruit}) = 1 - 0.75 = 0.25$$

$$n \text{ seedlings no fruit} = (0.25)^n$$

$$4) 99\% \text{ certain } P(\text{at least one}) = 0.99$$

$$\text{At least one} = 1 - \text{none}$$

$$(0.25)^n \leq 1 - 0.99$$

$$n \ln 0.25 \leq \ln 0.01$$

$$n \geq \frac{\ln 0.01}{\ln 0.25}$$

$$n \geq 3.32$$

$$\therefore n = 4$$

Q2

CSSA

There are 10 green marbles and w white marbles in a bag. The probability of selecting a white marble is $\frac{4}{9}$. How many more white marbles need to be added to the bag so that the probability of selecting a white marble from the bag is $\frac{3}{5}$?

(A) 7

(B) 8

(C) 14

(D) 21

Q3

CSSA

The local RSL is having a special promotion during 2015 where there will be a jackpot prize of \$5000 available at the end of each day. All patrons may enter only once each day. The prize is drawn daily and there is approximately a $\frac{1}{75}$ chance of the jackpot prize being won.

- What is the probability of a patron winning the jackpot prize on two consecutive days? 1
- How many consecutive draws must be made for the RSL to be 95% certain that a jackpot prize will be won? 3

(b) (i) (1 mark)

Sample answer:

$$P(\text{ww}) = \frac{1}{75} \times \frac{1}{75} = \left(\frac{1}{75}\right)^2 = \frac{1}{5625}$$

(b) (ii) (3 marks)

Sample answer:

$$P(\text{at least 1 jackpot from } n \text{ draws}) = 1 - P(\text{no jackpots from } n \text{ draws})$$

$$P(\text{no jackpots from 1 draw}) = 1 - \frac{1}{75} = \frac{74}{75} \quad P(\text{no jackpots from 2 draws}) = \frac{74}{75} \times \frac{74}{75} = \left(\frac{74}{75}\right)^2$$

$$P(\text{no jackpots from } n \text{ draws}) = \left(\frac{74}{75}\right)^n$$

For the probability of obtaining at least 1 jackpots from n draws to be 0.95:

$$1 - \left(\frac{74}{75}\right)^n = 0.95 \Rightarrow \left(\frac{74}{75}\right)^n = 0.05 \Rightarrow n \ln\left(\frac{74}{75}\right) = \ln 0.05$$

$$\therefore n = \frac{\ln 0.05}{\ln\left(\frac{74}{75}\right)} = 223.18 \quad \therefore n = 224 \text{ draws}$$



Probability

Part D: finding numbers



Q 4

FORT ST

The probability of Emily winning a particular game of chance is 30%.

- If she is to play 3 games, what is the probability she will win at least one game? **1**
- What is the probability she will win at least one game if she is to play 5 games? **1**
- What is the least number of games she must play before the probability of her winning at least one game is 99%? **2**

i. $1 - (0.7)^3 = 0.657$ **1**

ii. $1 - (0.7)^5 = 0.83193$ **1**

iii.

$$1 - (0.7)^n = 0.99$$

$$0.7^n = 0.01$$

$$n \log(0.7) = \log(0.01)$$

$$n = 13$$

1

Q5

FORT ST

Ava and Owen are playing a game by taking turns throwing a 20 sided die. Ava wins if she throws a number divisible by 5 or the number 13 and Owen wins if he throws a number divisible by 3.



Ava throws first.

- Find the probability that Ava wins the game on her first throw. **1**
- Find the probability that Ava wins the game on her first or second throw. **2**
- With how many of Ava's throws will she have had a better than a 50% chance of winning the game? **3**

(i) Solution

$$P(\text{Ava wins on 1st throw}) = \frac{4+1}{20} = \frac{1}{4}$$

(iii) Solution

$$P(\text{Ava wins}) =$$

$$\frac{1}{4} + \frac{1}{4} \left(\frac{21}{40} \right) + \frac{1}{4} \left(\frac{21}{40} \right)^2 + \frac{1}{4} \left(\frac{21}{40} \right)^3 + \dots + \frac{1}{4} \left(\frac{21}{40} \right)^n > 50\%$$

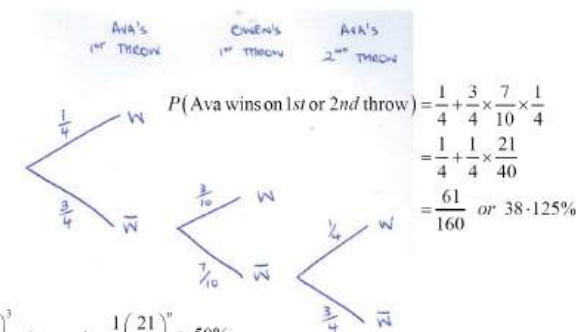
$$\frac{1 \left(\left(\frac{21}{40} \right)^n - 1 \right)}{\frac{21}{40} - 1} > 0.50$$

$$-\frac{10}{19} \left(\left(\frac{21}{40} \right)^n - 1 \right) > \frac{1}{2}$$

$$\left(\frac{21}{40} \right)^n - 1 < -\frac{19}{20}$$

$$\left(\frac{21}{40} \right)^n < \frac{1}{20}$$

(ii) Solution



So after 5 throws Ava would have had a better than 50% chance of winning the game.



Probability

Part D: finding numbers



2unit

selective

Q 6

GIRAWEEEN

A school sports team has a probability of 0.2 of winning any match.

- Find the probability the team wins exactly one of its first two matches. [2]
- Find the least number of consecutive matches the team must play to be 90% certain that it will win at least one match. [3]

$$\begin{aligned} \text{ii)} & P(WL) + P(LW) \\ &= 0.2 \times 0.8 + 0.8 \times 0.2 \\ &= 0.32 \end{aligned}$$

ii) let n be the number of matches needed.

$$\begin{aligned} \therefore P(\text{Win at least 1}) &= 1 - P(\text{all lose}) \\ &= 1 - 0.8^n \end{aligned}$$

$$\text{We need } 1 - 0.8^n = 0.9$$

$$\therefore 0.8^n = 0.1$$

$$n = \log_{0.8}(0.1)$$

$$n = \frac{\ln 0.1}{\ln 0.8} = 10.31 \dots$$

$\therefore$ need at least 11 games.

Q7

INDEPENDENT

A die is biased so that the probability of rolling a six is 0.1.

What is the minimum number of rolls of the die required to have a greater than 70% chance of rolling at least one 6?

Answer:

$$P(6) = 0.1$$

$$P(\bar{6}) = 0.9$$

$$\begin{aligned} P(\text{at least 1 "6"}) &= 1 - P(\text{all } \bar{6}) \\ &= 1 - (0.9)^n \end{aligned}$$

$$1 - (0.9)^n > 0.7$$

$$0.3 > (0.9)^n$$

$$\ln(0.3) > \ln(0.9)^n$$

$$\ln(0.3) > n \ln(0.9)$$

$$\frac{\ln(0.3)}{\ln(0.9)} < n$$

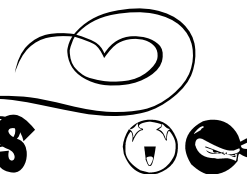
$$n > 11.42717266$$

$$n = 12$$



Probability

Part D: finding numbers



Q 8

INDEPENDENT

The probability that a rare plant grows to maturity in a nursery is $\frac{1}{9}$.

3

What is the least number of plants that would need to be in the nursery to have at least a 95% chance of one plant reaching maturity?

$$P(\text{at least 1}) = 1 - P(\text{all dead})$$

$$1 - \left(\frac{8}{9}\right)^n \geq 0.95$$

$$\left(\frac{8}{9}\right)^n \leq 0.05$$

$$\ln\left(\frac{8}{9}\right)^n \leq \ln(0.05)$$

$$n \ln\left(\frac{8}{9}\right) \leq \ln(0.05)$$

$$n \geq \ln(0.05) \div \ln\left(\frac{8}{9}\right)$$

$$n \geq 25.434..$$

$$n = 26$$

You need to have at least 26 plants have at least a 95% chance that one grows to maturity.

Q 9

INDEPENDENT

(i) A die is rolled 4 times. What is the probability that no sixes are rolled?

1

(ii) What is the minimum number of rolls required to ensure there is a 95% chance of getting a 6?

2

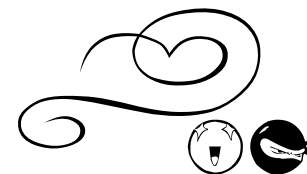
$$\begin{aligned} \text{(i)} \quad P(4 \times 6) &= \left(\frac{5}{6}\right)^4 \\ &= \frac{625}{1296} \end{aligned}$$

$$\begin{aligned} \text{(ii)} \quad 1 - \left(\frac{5}{6}\right)^n &> 0.95 \\ 0.05 &> \left(\frac{5}{6}\right)^n \\ \ln(0.05) &> n \ln\left(\frac{5}{6}\right) \\ n &> \ln(0.05) \div \ln\left(\frac{5}{6}\right) \\ n &> 16.43 \\ n &= 17 \end{aligned}$$



Probability

Part e: mixed



Q 1

Adelina and Tumaini are taking turns in a game with a single die, where the first person to roll a 6 is the winner. Adelina starts the game.

CSSA

- What is the probability that Adelina wins on her first turn? 1
- Draw a tree diagram with probabilities to indicate the outcomes of the first three turns of this game. 2
- What is the probability that Adelina eventually wins the game? 2

(b)(i) (1 mark)
Sample answer

$$P(\text{Adelina wins on her first turn}) = \frac{1}{6}$$

(b)(ii) (2 marks)

Sample answer



(b)(iii) (2 marks)

$$P(\text{Adelina eventually wins}) = \frac{1}{6} + \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \frac{5}{6} \times \frac{5}{6} \times \frac{5}{6} \times \frac{1}{6} + \dots$$

This is a limiting sum: $a = \frac{1}{6}$, $r = \frac{25}{36}$. Using $S_{\infty} = \frac{a}{1-r}$:

$$P(\text{Adelina eventually wins}) = \frac{\frac{1}{6}}{1 - \frac{25}{36}} = \frac{6}{11}$$

Q2

There are 4 red and 3 black discs in a bag. Sophie and Emma are playing a game in which they take turns drawing a disc from the bag and then replacing it.

CSSA

To win the game, Sophie must draw a red disc and for Emma to win she must draw a black disc. They continue taking turns until there is a winner. Sophie goes first.

- Find the probability that Sophie wins on her first draw. 1
- Find the probability that Sophie wins in three or less of her turns. 2
- Find the probability that Sophie wins the game. 2

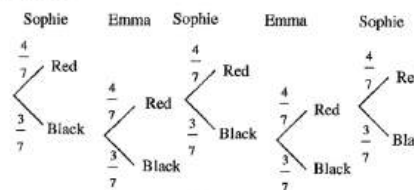
(b) (i) (1 mark)

Sample answer

$$P(\text{Sophie wins}) = \frac{4}{7}$$

(b) (ii) (2 marks)

Sample answer



$P(\text{Sophie wins in 3 or less turns})$

$$= P(\text{Sophie wins in 1 turn}) + P(\text{Sophie wins in 2 turns}) + P(\text{Sophie wins in 3 turns})$$

$$= \frac{4}{7} + \left(\frac{3}{7} \times \frac{4}{7} \times \frac{4}{7} \right) + \left(\frac{3}{7} \times \frac{4}{7} \times \frac{3}{7} \times \frac{4}{7} \right)$$

$$= \frac{4}{7} + \frac{48}{343} + \frac{576}{16807}$$

$$= 0.746$$

(b) (iii) (2 marks)

Sample answer

This is a limiting sum with $|r| < 1$, where $a = \frac{4}{7}$ and $r = \frac{12}{49}$.

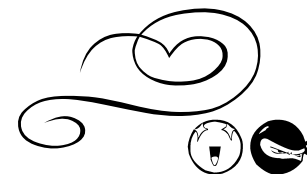
$$\text{So, } S_{\infty} = \frac{\frac{4}{7}}{1 - \frac{12}{49}} = \frac{28}{37}$$

So, the probability that Sophie wins the game is $\frac{28}{37}$.



Probability

Part e: mixed



Q3

GIRAWEEEN

Albert plays a game where he throws two standard six-sided dice and the total of the faces showing is noted. Albert wins the game if an 8 is thrown and he loses if a 5 is thrown. If the sum is any other number, the game continues until an 8 is thrown or a 5 is obtained.

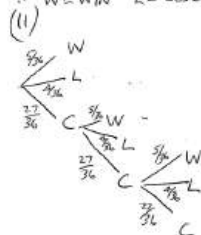
- Show that the probability that Albert wins on the first throw is $\frac{5}{36}$. 1
- Show that the probability that Albert wins on either the first, second or third throw is $\frac{185}{576}$. 2
- What is the probability that Albert wins the game? 2

(c)

1	2	3	4	5	6
1	2	3	4	5	6
2	3	4	5	6	7
3	4	5	6	7	8
4	5	6	7	8	9
5	6	7	8	9	10
6	7	8	9	10	11
6	7	8	9	10	11

$$P(8) = \frac{5}{36}$$

W = WIN L = LOSE C = CONTINUE



$$P = \frac{5}{36} + \frac{27}{36} \times \frac{5}{36} + \frac{27}{36} \times \frac{27}{36} \times \frac{5}{36}$$

$$= \frac{5}{36} \left(1 + \frac{27}{36} + \left(\frac{27}{36} \right)^2 \right)$$

$$= \frac{185}{576}$$

(iii) Continuing pattern above indefinitely

$$P = \frac{5}{36} \left(1 + \frac{27}{36} + \left(\frac{27}{36} \right)^2 + \dots \right)$$

LIMITING SUM

$$= \frac{5}{36} \left(\frac{1}{1 - \frac{27}{36}} \right)$$

$$= \frac{5}{36} \left(\frac{1}{1 - \frac{3}{4}} \right)$$

$$= \frac{5}{9}$$

Q4

SBHS

In a game, two players take turns at drawing and then immediately replacing a marble from a bag. The bag contains 2 green and 3 red marbles.

Player A draws first.

For A to win he must draw a green marble.

For B to win he must draw a red marble. Find the probability that

- A wins on his first draw. (1)
- B wins on his first draw. (1)
- A wins in fewer than 4 of his turns. (2)
- A wins eventually. (2)

(a) (i) $P(A \text{ wins on 1st draw})$

$$= \frac{2}{5}$$

(ii) $P(B \text{ wins on 1st draw})$

$$= \frac{3}{5} \times \frac{3}{5}$$

$$= \frac{9}{25}$$

(iii) $P(A \text{ wins in fewer than 4 turns})$

$$= P(\text{wins in 1}) + P(\text{wins in 2}) + P(\text{wins in 3})$$

$$= \frac{2}{5} + \frac{3}{5} \times \frac{2}{5} \times \frac{2}{5} + \frac{3}{5} \times \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5}$$

$$= \frac{2}{5} + \frac{12}{125} + \frac{72}{3125}$$

$$= \frac{1250 + 300 + 72}{3125}$$

$$= \frac{1622}{3125}$$

(iv) $P(A \text{ wins})$

$$= \frac{2}{5} + \frac{3}{5} \times \frac{2}{5} \times \frac{2}{5} + \frac{3}{5} \times \frac{2}{5} \times \frac{2}{5} \times \frac{2}{5} + \dots$$

$$= \frac{2}{5}$$

$$1 - \frac{6}{25}$$

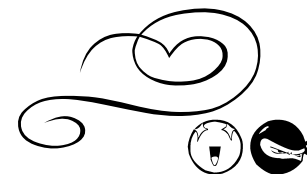
$$= \frac{19}{25}$$

$$= \frac{10}{19}$$



Probability

Part e: mixed



Q5

SBGHS

The faces of a ten-sided die are labelled with the numbers 1 to 10 and all faces are equally likely to appear when the die is rolled.

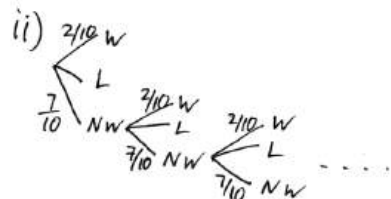
Dianne plays a game by rolling the ten-sided die until she wins or loses the game.

- If she rolls a 9 or 10 at any time she wins the game.
- If she rolls a 1 at any time she loses the game.

Otherwise, she must keep rolling the die.

- (i) What is the probability that Dianne rolls the die less than three times? 2
- (ii) What is the probability that Dianne wins the game at any time? 2

$$\begin{aligned} \text{e) i) } & \frac{2}{10} + \frac{1}{10} + \frac{7}{10} \left(\frac{2}{10} + \frac{1}{10} \right) \checkmark \\ & = \frac{51}{100} \checkmark \end{aligned}$$



$$P(\text{win}) = \frac{2}{10} + \frac{7}{10} \times \frac{2}{10} + \left(\frac{7}{10} \right)^2 \times \frac{2}{10} + \left(\frac{7}{10} \right)^3 \times \frac{2}{10} \dots \checkmark$$

This is a Geo Series

$$S_{\infty} = \frac{a}{1-r} = \frac{2/10}{1-7/10} = \frac{2}{3} \checkmark$$

Q4

SBHS